

j4_controller

Firmware for the **Johnny 4 robot controller/transmitter**. Runs on a LILYGO TTGO T-Display v1.1 (ESP32). Reads potentiometers and two 4x4 keypads, ingests four more pots from the j4_display_right board, transmits control data to the robot receiver over ESP-NOW, and mirrors live data to the two XIAO display boards over UART.

What is Johnny 4?

Johnny 4 is a prop robot controlled wirelessly. This board is the handheld controller that the operator uses to drive the robot's servos, select audio phrases, and monitor battery status.

What this firmware does

- Reads analog potentiometers via four local ADS1115 ADC modules: the eyes joystick (eyes_x / eyes_y), the neck joystick (neck X / jaw Y), and the eight middle face pots (Eyebrow L/R, Basket Eyebrow L/R, Nose, Nose Basket, Bottom Eyelid L/R); the fourth module carries the neck-pivot pot, the two linear faders, and the Nose Basket pot
- Reads the four panel toggle switches (LASER, VENT, EYE POP, and the AUX toggle by the right joystick) on GPIOs 32/33/13/15 with internal pull-ups
- Ingests four more pots (iris, color, brightness, volume) from [j4_display_right](#)'s dedicated ADS1115 over Serial2 as 25 Hz P: lines
- Reads two 4x4 matrix keypads via PCF8574 I2C expanders using a custom scan routine
- Left keypad: jukebox-style audio phrase selection: press a letter (A-D) then a digit (0-9) to queue a phrase, or press a digit alone to queue a 0x phrase (e.g. pressing 8 queues "08.wav")
- Right keypad: face presets. Tap a key to recall its saved face; hold a key 3 seconds to save the current face to it (j4_display_right shows a confirm prompt, * confirms). Faces persist on j4_talk's microSD and are re-loaded in the background whenever the talk link is up
- Press * to send a STOP command, press # to clear the buffer
- Mixes the joystick into differential neck-left / neck-right values over a 0-3200 range. Both axes are sticky-filtered at the source so a parked (spring-removed) stick holds dead still without giving up 1-count precision when moved slowly
- Transmits all control data to the robot receiver via ESP-NOW (fixed-size packed structs, no String members so they survive the wireless memcpy)
- Receives the jukebox file list from the robot receiver in chunks and forwards it to the display board; carries a need_filelist flag so the receiver re-sends the list if this board boots late
- Answers LIST? requests from the display board out of its cached copy, so a display reboot recovers the list without a full round trip
- Displays live pot values, keypress state, and battery voltage on the built-in TFT. The default (data) screen reads: Keypad_L, Playing, VOL, Keypad_R, Eye-X, Eye-Y, Neck-L, Neck-R, Neck-PIV, then the STATUS line. Each keypad has its own last-key line
- The TTGO's built-in button (GPIO 35) cycles the TFT through seven screens: live data, WiFi MAC address, a connection-status screen showing ESP-NOW LINK, j4_stepper_neck, j4_stepper_eyes, j4_talk, j4_display_left, and j4_display_right as CONNECTED (green) / DISCONNECTED (red), then four more showing each ADS1115 module's live raw pot counts (ADS_01 through ADS_04) for bench testing without a laptop on the I2C bus
- Sends a 49-byte binary status packet to the j4_display_left board at 25 fps over UART
- Shows a STATUS line on the built-in TFT: "ONLINE" when the ESP-NOW link is up and the steppers are healthy, "OFFLINE" if no status packet arrives, or the reported stepper fault (e.g. "NL OT", "EYES OFFLINE"); green when healthy, red on any fault

Hardware

Component	Details
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Microcontroller	LILYGO TTGO T-Display v1.1 (ESP32 with built-in 1.14" ST7789 TFT)
ADC modules	Four ADS1115 (I2C addresses 0x48, 0x49, 0x4A, 0x4B); a fifth lives on j4_display_right
Toggle switches	LASER, VENT, EYE POP, AUX -- GPIOs 32/33/13/15, INPUT_PULLUP, switch closes to GND
Keypad expanders	Two PCF8574 I2C I/O expanders: 0x21 (keypad_left, A0 jumper high) and 0x20 (keypad_right)
Keypads	Two 4x4 matrix keypads, headers soldered directly into P0-P7; left = phrase select, right = face presets
Displays	j4_display_left (jukebox, Serial1) + j4_display_right (pot labels, Serial2)

Pin assignments

GPIO	Function
21	I2C SDA (ADS1115 x4, PCF8574 keypads x2)
22	I2C SCL
17	Serial1 TX to j4_display_left (XIAO D7)
27	Serial1 RX from j4_display_left (XIAO D6)
25	Serial2 TX to j4_display_right (XIAO D7, face-preset messages)
26	Serial2 RX from j4_display_right (XIAO D6, pot feed)
32	LASER toggle (INPUT_PULLUP, switch closes to GND)
33	VENT toggle (INPUT_PULLUP, switch closes to GND)
13	EYE POP toggle (INPUT_PULLUP, switch closes to GND)
15	AUX toggle, next to the right joystick (INPUT_PULLUP, unassigned)
34	Battery voltage sense (ADC input, read-only)
35	Screen-cycle button (TTGO built-in; data / MAC / connection status)

GPIO 35 is the TTGO's on-board button, so it needs no wiring. Potentiometer inputs are routed through the ADS1115 modules. GPIO 12 causes boot/flash issues when connected to a pot and is avoided. GPIO 17 is unreliable as an input and is used only as TX. Pins 36, 37, and 38 are input-only. The four toggles use the internal pull-ups, so each switch just shorts its GPIO to GND when flipped ON -- no external resistors.

Pin diagram

Physical layout, display facing AWAY from you (rails visible), USB-C at the TOP (matching how the board is mounted). The two header rails read top-to-bottom as shown.

LEFT rail		RIGHT rail
	+- - [USB-C] - - +	
3.3V -- 3V3		5V
ground -- GND		GND ground

```

        ground -- GND|      TTGO      | 27  DISP-L RX <- XIA0 D6
(avoid: boot/flash) -- 12 | T-Display | 26  DISP-R RX <- XIA0 D6
        EYE POP toggle -- 13 | v1.1   | 25  DISP-R TX (face msgs)
        AUX toggle -- 15 |         | 33  VENT toggle
        (free) -- 2 | [ ST7789 | 32  LASER toggle
DISP-L TX -> XIA0 D7 -- 17 | TFT on  | 39  (input only)
        I2C SCL -- 22 | back ]  | 38  (input only)
        I2C SDA -- 21 |         | 37  (input only)
        ground -- GND|         | 36  (input only)
        ground -- GND|         | 3.3V
                        +-----+

```

```

on-board (no header pin): GPIO34 = battery sense, B00T = GPIO0,
GPIO35 = screen-cycle button, TFT on 4/5/16/18/19/23
I2C bus: ADS_01 @ 0x48, ADS_02 @ 0x49, ADS_03 @ 0x4A, ADS_04 @ 0x4B,
PCF8574 keypad_left @ 0x21, keypad_right @ 0x20
toggles: INPUT_PULLUP, switch closes to GND (ON = LOW)

```

Corrected against LilyGO's own pinout diagram for the T-Display v1.1 (24 pins, 12 per rail, drawn from the front with the display facing the viewer and USB-C at the bottom). Getting from that diagram to this one takes two transforms, not one: rotate 180 degrees so USB-C ends up at the top (this alone swaps which rail each pin is on *and* reverses top-to-bottom order), then mirror left-right because this diagram is drawn from the back (rails visible, display facing away) rather than the front (a plain left/right swap, order unchanged). Composed, the right rail happens to end up identical to where it started; the left rail does not. The previous version of this diagram had the left rail's pins in the un-rotated order and was missing a GND pin on each rail (11 shown per side against the board's actual 12) -- both are fixed above.

ESP-NOW (wireless) links this board to j4_receiver.

ADS1115 pin diagrams

All four modules share the TTGO's I2C bus (SDA = GPIO 21, SCL = GPIO 22, 3.3V, GND); only the ADDR strap and the four analog channels differ. Every pot's wiper goes to its A-pin; the outer legs go to 3.3V and GND.

```

ADS_01 @ 0x48 (ADDR -> GND)          both joysticks
+-----+
| VDD  | 3.3V
| GND  | GND
| SCL  | TTGO GPIO 22 (I2C SCL)
| SDA  | TTGO GPIO 21 (I2C SDA)
| ADDR | GND = address 0x48
| ALRT | not connected
| A0   | neck joystick X wiper      -> neck L/R differential mix
| A1   | neck joystick Y (jaw) wiper -> neck L/R base height
| A2   | eyes joystick X wiper      -> eyes pan servo (PCA9685 ch 3)
| A3   | eyes joystick Y wiper      -> eyes tilt servo (PCA9685 ch 4)
+-----+

```

```

ADS_02 @ 0x49 (ADDR -> VDD)          four face pots, left bank
+-----+
| VDD  | 3.3V
| GND  | GND
| SCL  | TTGO GPIO 22 (I2C SCL)
| SDA  | TTGO GPIO 21 (I2C SDA)

```

```

| ADDR | VDD = address 0x49
| ALRT | not connected
| A0 | Eyebrow L pot wiper -> PCA9685 ch 6
| A1 | Eyebrow R pot wiper -> PCA9685 ch 7
| A2 | Basket Eyebrow L pot wiper -> PCA9685 ch 8
| A3 | Basket Eyebrow R pot wiper -> PCA9685 ch 9
+-----+

ADS_03 @ 0x4A (ADDR -> SDA) three face pots, right bank
+-----+
| VDD | 3.3V
| GND | GND
| SCL | TTGO GPIO 22 (I2C SCL)
| SDA | TTGO GPIO 21 (I2C SDA)
| ADDR | SDA = address 0x4A
| ALRT | not connected
| A0 | Nose pot wiper (up/down) -> PCA9685 ch 10
| A1 | FAULTY channel, nothing connected, left unread in firmware
| A2 | Bottom Eyelid L pot wiper -> PCA9685 ch 12
| A3 | Bottom Eyelid R pot wiper -> PCA9685 ch 13
+-----+

ADS_04 @ 0x4B (ADDR -> SCL) neck-pivot + faders + Nose Basket
+-----+
| VDD | 3.3V
| GND | GND
| SCL | TTGO GPIO 22 (I2C SCL)
| SDA | TTGO GPIO 21 (I2C SDA)
| ADDR | SCL = address 0x4B
| ALRT | not connected
| A0 | neck-pivot pot wiper -> nP on the stepper link (-1600..0..1600)
| A1 | fader_left wiper -> neck pivot (doubles ADS_04 A0, inverted)
| A2 | fader_right wiper -> iris (doubles j4_display_right IRIS)
| A3 | Nose Basket pot wiper -> PCA9685 ch 11 (single source)
+-----+

```

Analog inputs

Local, on the four ADS1115 modules over I2C:

Module	Channel	Control
ADS_01 (0x48)	A0	Neck joystick X
ADS_01 (0x48)	A1	Neck joystick Y (jaw, feeds neck mixer)
ADS_01 (0x48)	A2	Eyes joystick X / eyes_x
ADS_01 (0x48)	A3	Eyes joystick Y / eyes_y
ADS_02 (0x49)	A0	Eyebrow L pot
ADS_02 (0x49)	A1	Eyebrow R pot

ADS_02 (0x49)	A2	Basket Eyebrow L pot (was eye-pop, now a toggle)
ADS_02 (0x49)	A3	Basket Eyebrow R pot
ADS_03 (0x4A)	A0	Nose pot (up/down)
ADS_03 (0x4A)	A1	Faulty channel, unused (Nose Basket moved to ADS_04 A3)
ADS_03 (0x4A)	A2	Bottom Eyelid L pot
ADS_03 (0x4A)	A3	Bottom Eyelid R pot
ADS_04 (0x4B)	A0	Neck-pivot pot (silver knob below j4_display_left, -1600..0..1600)
ADS_04 (0x4B)	A1	fader_left (linear fader, doubles the neck-pivot pot, same scale, read inverted)
ADS_04 (0x4B)	A2	fader_right (linear fader, doubles the IRIS pot)
ADS_04 (0x4B)	A3	Nose Basket pot (up/down, single source)

The eight face pots ride the ESP-NOW control packet to PCA9685 channels 6-13 on j4_receiver (Eyebrow L/R = ch 6/7, Basket Eyebrow L/R = ch 8/9, Nose = ch 10, Nose Basket = ch 11, Bottom Eyelid L/R = ch 12/13). The neck-pivot pot rides the packet in its own field and leaves the receiver as the nP slot of the stepper stream.

Centre-zero controls and jitter filtering

The neck-pivot pot and fader_left are read centre-zero: **-1600 at minimum, 0 at the midpoint, +1600 at maximum**. fader_left is read inverted so that raising the fader and turning the pot drive the pivot the same direction. Everything else stays on its unsigned scale.

processPotCentred() filters these two, because an unfiltered ADC value wanders a few counts even when nothing is touched and every wander that survives the downstream threshold is a real motor step:

- **Deadband** (POT_DEADBAND , 30 counts): anything within this of centre reads exactly 0, so a control at rest commands a hard stop rather than a small standing offset.
- **Sticky band** (POT_STICKY_BAND , 3 counts): see below. Entering the deadband always snaps to 0, so coming to rest never strands a residual.

Sticky band: stillness at rest without losing resolution

stickyBand() is the anti-jitter filter used by the neck joystick axes and, inside processPotCentred(), by the neck pivot and fader_left.

The obvious filter -- "only report a new value once it has moved N counts, then jump to it" -- does stop jitter, but it also quantises real movement into N-sized steps, so inching a control makes the output hop N at a time instead of counting. That is unusable on an axis you want to move slowly and precisely.

Instead the reported value **trails** the live one by up to band and is dragged along rather than snapped:

- **At rest:** noise inside +/- band never moves the reported value at all.
- **Moving:** the report follows the input one count at a time, staying band behind, so full 1-count resolution survives.

The only cost is band counts of lag, under a tenth of a percent of a 3200-count axis.

The **neck joystick axes are filtered at the two axis reads, before the mixer**. neck_left and neck_right are a differential mix of both axes, so a couple of counts of noise on each reaches the output as a couple of counts of physical

stepper motion. Filtering the mixer outputs instead would leave each axis free to jitter into the other. This matters most with the joystick springs removed, when the stick can sit parked off-centre indefinitely.

Simulated against +/-8 raw counts of ADC noise with the stick parked off-centre: `neck_left` changes value **3990 times in 5000 reads unfiltered, and 0 times filtered**. A slow deliberate sweep still resolves 110 distinct values across a 111-count span, so the precision is intact.

Unlike `processPot()`, this does not treat a low reading as a noise floor. That guard is free on an unsigned scale where 0 is also the bottom of travel, but here 0 is the centre, so it would make a pot turned fully to minimum snap to centre and lose the bottom of its travel.

The ESP-NOW field stays 0-3200 absolute and is re-centred at transmit. `j4_stepper_neck` turns `nP` into an absolute position (`nP/2`) homed off the MIN limit switch, so signed values on the wire would command negative positions and run the pivot into that switch. Keeping the wire contract also means `j4_receiver` and `j4_stepper_neck` need no reflash to match this build.

The two linear faders each double an existing rotary pot, arbitrated last-mover-wins: whichever control of the pair moved last (beyond a small claim threshold) is the active source and its value is used, so the two never fight. `fader_right` pairs with the IRIS pot on `j4_display_right`; `fader_left` pairs with the neck-pivot pot. A source that is absent (module unplugged, display feed down) can neither claim nor hold active status.

The claim threshold is per-pair rather than a single global constant, because the pairs are not on the same scale: the iris pair runs 0-255 and the neck-pivot pair spans 3200 counts (-1600..1600). A threshold has to be a similar fraction of travel on each (about 1.5%), so it is 4 counts for iris and 50 for neck pivot. A single 4-count threshold would be 0.125% of neck-pivot travel, inside the pots' own noise, and the pair would flip-flop between sources on noise alone.

Both halves of the neck-pivot pair (rotary pot on A0, fader on A1) live on ADS_04, so losing that one module drops both sources at once; the value falls back to the 1600 centre rather than to a hard-over endpoint.

Remote, streamed from `j4_display_right`'s dedicated ADS1115 (raw counts over Serial2, scaled here with the same `processPot()`):

Channel	Control	Destination
A0	Iris	270-deg iris servo (PCA9685 on <code>j4_receiver</code>)
A1	Color	WS2812B strip color (<code>j4_receiver</code>)
A2	Brightness	WS2812B strip brightness (<code>j4_receiver</code>)
A3	Volume	<code>j4_talk</code> audio volume

`eyes_x`, `eyes_y`, and iris drive servos on the receiver's PCA9685. The neck joystick mixes into neck-L / neck-R as before. color and brightness drive the receiver's WS2812B strip.

Toggle switches

Four panel toggles, each wired between its GPIO and GND (internal pull-up, ON = LOW). Their states travel in the control packet as a bitmask:

GPIO	Toggle	Bit	Action on the robot
32	LASER	0	Laser servo, PCA9685 ch 14 on <code>j4_receiver</code>
33	VENT	1	Vent-fin servo, PCA9685 ch 15 on <code>j4_receiver</code>

13	EYE POP	2	Eye-pop steppers: sends 0 (normal) or 3200 (popped) down the stepper chain (j4_stepper_neck to j4_stepper_eyes)
15	AUX	3	Next to the right joystick; read and transmitted, not yet assigned

Keypad wiring

Two 4x4 keypads, each on its own PCF8574 backpack on the shared I2C bus:

- **keypad_left** (the newer keypad, left of the panel) at **0x21** -- set the backpack's A0 jumper high. If that backpack turns out to be a PCF8574A, the same jumper lands at 0x39; change the two `0x21`s in `main.cpp`.
- **keypad_right** (the original keypad) at **0x20** -- all jumpers low.

The left keypad drives phrase select; the right keypad drives face presets (tap = recall, 3s hold = save prompt, * confirms). Each can be absent or hot-plugged without affecting the other.

Each keypad is plugged straight into its PCF8574 P0-P7 (keypad pin 1 into P0, in order). The I2CKeyPad library's fixed scan pattern does not match this wiring, so the firmware uses a custom scan via the PCF8574 library directly.

The two pads are different models and **do not put their rows and columns on the same pins**, so both the keymap string and the drive/read pin split are per-keypad (`KeypadPins` , carried in `KeypadGuard`).

Right keypad (original) -- rows on P0, P1, P2, P7; scanner drives those and reads P3, P4, P5, P6:

PCF pin	Keypad wire
P0	Column 3 (3, 6, 9, #)
P1	Column 2 (2, 5, 8, 0)
P2	Column 1 (1, 4, 7, *)
P3	Row 4 (*, 0, #, D)
P4	Row 3 (7, 8, 9, C)
P5	Row 2 (4, 5, 6, B)
P6	Row 1 (1, 2, 3, A)
P7	Column 4 (A, B, C, D)

Left keypad (newer model) -- straight 4+4 split; scanner drives P0-P3 and reads P4-P7:

PCF pin	Keypad wire
P0	Column 1 (1, 4, 7, *)
P1	Column 2 (2, 5, 8, 0)
P2	Column 3 (3, 6, 9, #)
P3	Column 4 (A, B, C, D)
P4	Row 1 (1, 2, 3, A)
P5	Row 2 (4, 5, 6, B)

P6	Row 3 (7, 8, 9, C)
P7	Row 4 (*, 0, #, D)

Why the split matters (bench note, 2026-08-29)

A matrix key is only visible to the scanner if one of its two lines is in the drive set and the other is in the read set. Running the left pad with the right pad's split left six keys (A, B, C, *, 0, #) with both of their lines in the same set, so they produced no scan code at all and the remaining ten came out transposed. No keymap string can recover keys that never generate a code, which is why this needed the per-pad `KeypadPins` change rather than just a re-derived `keymap_left`.

To re-derive a keymap for a replacement pad: press each key, note the character that appears, and rearrange that pad's string so index `drive*4 + read` holds the right character. If some keys produce nothing at all, the drive/read split is wrong, not the string.

Face presets

A **face** is a snapshot of the robot's facial expression: iris, eye color, eye brightness, the eight middle face pots, and the LASER / VENT / EYE POP toggle states. Volume, the neck values, and eyes X/Y are deliberately not part of a face.

On the **right keypad**:

- **Tap a key**: recall that key's saved face. Every face channel jumps to the saved value and holds there; turning any face pot takes just that channel back (frozen-baseline takeover, so even a slow turn works), and flipping a toggle takes that toggle back. The rest of the face stays put.
- **Hold a key for 3 seconds** (any key except `*`): `j4_display_right` shows "SAVE FACE ON `<key>` ? PRESS * TO CONFIRM". `*` saves, any other key cancels, and an unanswered prompt times out after 10 seconds.

Faces persist in `FACES.TXT` on `j4_talk`'s microSD, so they survive power-off. The sync is fully background and never blocks anything:

- After boot (or whenever the talk link appears) this board re-requests the saved-face dump every 2.5s until a complete one lands.
- A save made while `j4_talk` is offline stays in RAM flagged dirty ("PENDING SD" on the display) and is pushed automatically when the link comes up; the display shows "FACE `<key>` ON SD" when the Teensy confirms the write.
- If the Teensy reports an SD write failure, the face stays usable in RAM for the session and auto-retry stops until it is re-saved.

Slots are keyed by the keypad **character**, so re-deriving a keymap keeps every saved face on the same printed key. 15 keys are usable as slots (`*` is the confirm key). Face traffic is its own ESP-NOW packet type (0x04) translated by `j4_receiver` to text lines on the Teensy UART; see those repos for the protocol.

Dependencies

Managed via PlatformIO (`platformio.ini`):

Library	Version
RobTillaart/PCF8574	0.3.9
RobTillaart/ADS1X15	0.3.13
TFT_eSPI	bundled in <code>lib/</code> (pre-configured for TTGO T-Display v1.1)

Built-in ESP-IDF components used: `esp_now` , `WiFi` .

Building and uploading

```
# Build
pio run

# Upload
pio run --target upload

# Monitor serial output
pio device monitor
```

Upload speed is 921600 baud. Serial monitor is 115200 baud. Both are set in `platformio.ini` .

The TFT_eSPI copy in `lib/TFT_eSPI/` is pre-configured for the TTGO T-Display v1.1. Do not replace it with the registry version without reconfiguring.

Receiver

The robot receiver's MAC address is set in `broadcastAddress[]` near the top of `main.cpp` . Update this if the receiver board is swapped.

Related projects

- [j4_display_left](#) -- the XIAO ESP32S3 jukebox display (portrait, left of the panel)
- [j4_display_right](#) -- the XIAO ESP32S3 pot-label display (landscape, right of the panel) that streams the iris/color/brightness/volume pots to this board
- [j4_receiver](#) -- the robot-side TTGO that receives this controller's ESP-NOW packets and returns status